

EDS Data Discrepancy & Audit Report

Reconciliation of EDS_Full Data Table Consolidation_30 Aug.xlsx vs
EDS_Cleaned_Master_2July.xlsx

EXECUTIVE SUMMARY

An exhaustive cell-by-cell and column-by-column audit was conducted comparing the 30 August Consolidation workbook with the 2 July Cleaned Master dataset. While key headline statistics (digital completion: 86.7%, IPT attendance: 93.3%, CLSS attendance: 88.3%) are consistent, critical structural flaws, missingness gaps, and logical row contradictions were identified in the master dataset.

1. Sample Size & Export Artifacts

The raw **Quant_Structured** sheet contains **75 rows** in Excel, creating a potential calculation baseline error if computed naively:

Sheet Name	Raw Rows	Header Rows	Trailing Blank Rows	Valid Sample (N)
Quant_Structured	75	1 (Row 0)	14 (Rows 61–74)	60 Teachers
Qual_FreeText	61	1 (Row 0)	0	60 Teachers
Obs_Observations	61	1 (Row 0)	0	60 Teachers

2. Internal Logical Contradictions in July Master File

- **Teacher 38 CLSS Participation Contradiction:** Col 18 (*CLSS Flag*) is marked **1** (Participated), but Col 68 (*No. of CLSS Attended*) explicitly records **'Not Participated'**. This causes an attendee sum mismatch (53 vs 52).
- **Online Course Derivative Flag Contradiction:** Teachers 18, 26, 27, 28, 34, 44, 47, and 49 answered *No* for completing online courses (Col 37), but Col 96 records string **'0'** instead of remaining null.
- **Duplicate Header Rows:** Row 0 across Quant_Structured, Qual_FreeText, and Obs_Observations repeats string column headers as data values (*Teacher_ID* = *'Teacher_ID'*).
- **Uncleaned Scratch Sheets:** Sheet1 contains misspelled headers (*'Annual Enagement '*) and Sheet2 contains raw unformatted pivot outputs.

3. High Field Missingness (Implicit vs Explicit Nulls)

High missingness in binary indicator columns indicates that enumerators skipped checkboxes when respondents replied 'No' or 'Unaware', rather than explicitly entering 0 or No:

Column Name & Question	Total N	Non-Null Count	Missing (NaN)	Missing %
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Quant Col 62: Knowledge of DIKSHA	60	23 Yes	37 missing	61.7%
Quant Col 87: Knowledge of CLSS Format	60	10 Yes, 3 No	47 missing	78.3%
Quant Col 33: Classroom Resource Application	60	38 answered	22 missing	36.7%
Obs: Material Access & Adoption Notes	60	32 answered	28 missing	46.7%

4. Cell-by-Cell Metric Reconciliation Matrix

Reconciliation of all key metrics in 30 Aug Consolidation against July Master dataset:

Domain / Metric	30 Aug Claim	July Master Re-Calculated	Denominator	Status
Digital Course Completion	52 / 60 (87%)	52 / 60 (86.7%)	N = 60	Match
Timing: After School	48 / 60 (80%)	48 / 60 (80.0%)	N = 60	Match
Timing: During School	22 / 60 (37%)	22 / 60 (36.7%)	N = 60	Match
Subject Training Attendance	56 / 60 (93%)	56 / 60 (93.3%)	N = 60	Match
Training Modules Received	36 teachers	36 / 60 (60.0%)	N = 60	Match
IPT Materials: None Received	19 teachers	19 / 56 attendees (33.9%)	N = 56	Match
CLSS Attendance (1+ Session)	53 / 60 (88%)	53 / 60 (88.3%)	N = 60	Match
Attended All 5 CLSS Sessions	20 / 53 (38%)	20 / 53 (37.7%)	N = 53	Match
CLSS Solved Challenge	32 / 60 (53%)	32 / 60 (53.3%)	N = 60	Match
CLSS Topic Non-Recall	18 reported	35 / 53 (66.0%)	N = 53	Recalibrated

5. Multi-Selection Column Totals & Denominators

- **DIKSHA Engaging Features (Cols 43–49):** 86 selections across 40 responding teachers (20 left section blank). Calculating % over N=40 (quizzes: 50.0%) vs N=60 (quizzes: 33.3%) creates a **16.7 percentage point mismatch**.
- **CLSS Attendance Form Friction (Cols 82–86):** 66 selections across 53 CLSS attendees. Calculating % over N=53 vs N=60 creates a **7.5 percentage point shift**.
- **CLSS Topic Non-Recall Deficit (Cols 69–74):** 18 teachers explicitly marked 'No recall', while 17 attendees left all topic checkboxes blank. Combined non-recall is **35 of 53 attendees (66.0%)**.

RECOMMENDED CLEANING ACTIONS FOR JULY MASTER DATASET

1. **Strip Trailing Rows:** Delete blank export rows 61–74 in Quant_Structured and drop string header row 0.
2. **Resolve Teacher 38 Contradiction:** Reconcile CLSS flag (Col 18) and attendance count (Col 68).
3. **Standardize Null Values:** Convert missing NaN values in binary awareness columns (Cols 62, 87) to explicit 0 (No).
4. **Purge Scratch Sheets:** Remove uncleaned Sheet1 and Sheet2.